

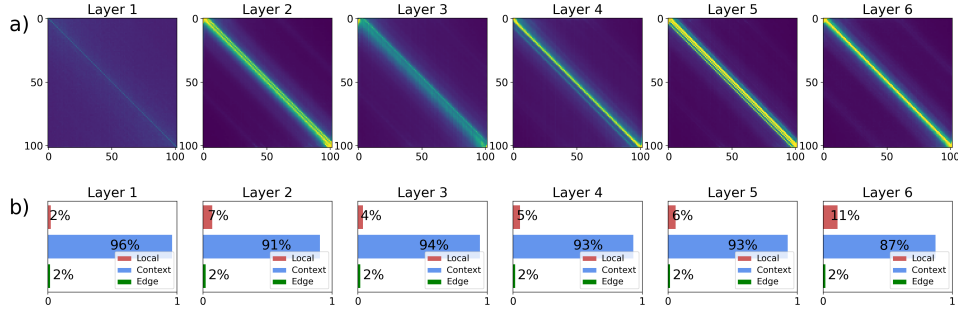


Figure 6: Attention Map Analysis for MLM-S model. **a)** Average attention logits per layer, aggregated over all heads and 128 protein sequences, each consisting of 100 amino acids. **b)** Distribution of attention scores across different types: *Context* attention, *Local* attention, and edge-token attention.

We can better understand which elements influence a given token’s representation and how contextual information is integrated by analyzing the attention map of the transformer model and studying its distributions across the layers

In figures 5 and 6, we define *Context* as the sum of the attention logits that connect each position in the sequence to all other different positions. *Local* refers to the attention logits located along the diagonal of the attention weight matrix, representing how much a position attends to itself. Lastly, *Edge* corresponds to the attention logits assigned to the EOS and BOS tokens.

For the LSD-NM-S diffusion model, the early layers predominantly focus on contextual information, with *Context* attention reaching 95% and *Local* attention remaining as low as 1%. This suggests that initial layers are primarily responsible for embedding global context into each amino acid position. As the layer index increases, attention shifts to be more *Local* focused, indicating that final layers refine token embeddings based on the already incorporated contextual information. The model exhibits minimal focus on edge tokens, with attention weight not exceeding 3% and dropping to 0% in the final layer.

In contrast, the MLM-S model maintains a strong reliance on context across all layers, consistently prioritizing *Context* attention. A key difference is that attention in the MLM-S model is more short-range, with logits concentrated around nearby positions along the diagonal, whereas LSD-NM-S diffusion model distributes attention more broadly. This distinction highlights the differing information integration strategies between the two models, where LSD-NM-S diffusion model gradually transitions from global to local representation, while MLM-S persistently relies on short-range context.